

Penicillin and Cephalosporin cross-reactivity. Beta-Lactam antibiotics can have similar or identical R1 or R2 side chains, which may make cross-reactivity more likely. This matrix indicates either a similar (gray) or an identical (red) side chain. Empty boxes indicate a lack of side-chain similarity.

	Cefazolin (1 st)	Cefaclor (2 nd)	Cefadroxil (1 st)	Cefamandole(2 nd)	Cefdinir (3 rd)	Cefepime (4 th)	Cefixime (3 rd)	Cefoperazone (3 rd)	Cefotaxime (3 rd)	Cefotetan (2 nd)	Cefoxitin(2 nd)	Cefpirome(4 th)	Cefpodoxime (3 rd)	Cefprozil (2 nd)	Ceftazidime (3 rd)	Ceftolozane (2nd)	Ceftibuten (3 rd)	Ceftizoxime (3 rd)	Ceftriaxone (3 rd)	Cefuroxime (2 nd)	Cephalexin (1 st)	Cephaloridine (1 st)	Cephadrine (1 st)	Cefditoren (3 rd)	Ceftaroline (5 th)	Amoxicillin	Ampicillin	Penicillin G	Aztreonam
Cefazolin (1 st)	-																												
Cefaclor (2 nd)		-	☒	☒									☒								☒	☒				☒	☒		
Cefadroxil (1 st)		☒	-	☒									☒								☒	☒				☒	☒		
Cefamandole (2 nd)		☒	☒	-				☒		☒				☒							☒	☒				☒	☒		
Cefdinir (3 rd)					-	☒																							
Cefepime (4 th)						-	☒		☒			☒	☒		☒	☒		☒	☒	☒				☒					☒
Cefixime (3 rd)					☒	☒	-		☒			☒	☒		☒	☒		☒	☒	☒				☒					☒
Cefoperazone (3 rd)				☒				-		☒																			
Cefotaxime (3 rd)						☒	☒		-			☒	☒		☒	☒		☒	☒	☒				☒					☒
Cefotetan (2 nd)				☒				☒		-																			
Cefoxitin(2 nd)											-									☒		☒							☒
Cefpirome(4 th)						☒	☒		☒			-	☒		☒	☒		☒	☒	☒				☒					☒
Cefpodoxime (3 rd)						☒	☒		☒		☒	-			☒	☒		☒	☒	☒				☒					☒
Cefprozil(2 nd)		☒	☒	☒									-								☒		☒			☒	☒		
Ceftazidime (3 rd)						☒	☒		☒			☒	☒		-	☒		☒	☒	☒				☒					☒
Ceftolozane (2nd)						☒	☒		☒			☒	☒		☒	-		☒	☒	☒				☒					☒
Ceftibuten (3 rd)																-													
Ceftizoxime (3 rd)						☒	☒		☒			☒	☒		☒	☒	☒	-	☒	☒				☒					☒
Ceftriaxone (3 rd)						☒	☒		☒			☒	☒		☒	☒	☒		-	☒				☒					☒
Cefuroxime(2 nd)						☒	☒		☒		☒	☒	☒		☒	☒	☒	☒		-			☒						☒
Cephalexin (1 st)		☒	☒	☒										☒							-	☒				☒	☒		
Cephaloridine (1 st)										☒												-							☒
Cephadrine (1 st)		☒	☒	☒										☒							☒		-			☒	☒		
Cefditoren (3 rd)						☒	☒		☒			☒	☒		☒	☒	☒	☒					-						☒
Ceftaroline (5 th)																								-					
Amoxicillin		☒	☒	☒									☒								☒		☒			-	☒		
Ampicillin		☒	☒	☒									☒								☒	☒			☒		-		
Penicillin G											☒											☒							-
Aztreonam						☒	☒		☒			☒	☒		☒	☒		☒	☒	☒				☒					-

Source: Ana Dioun Broyles, Aleena Banerji, Sara Barmettler, Catherine M. Biggs, Kimberly Blumenthal, Patrick J. Brennan, Rebecca G. Breslow, Knut Brockow, Kathleen M. Buchheit, Katherine N. Cahill, Josefina Cernadas, Anca Mirela Chiriac, Elena Crestani, Pascal Demoly, Pascale Dewachter, Meredith Dilley, Jocelyn R. Farmer, Dinah Foer, Ari J. Fried, Sarah L. Garon, Matthew P. Giannetti, David L. Hepner, David I. Hong, Joyce T. Hsu, Parul H. Kothari, Timothy Kyin, Timothy Lax, Min Jung Lee, Kathleen Lee-Sarwar, Anne Liu, Stephanie Logsdon, Margee Louisias, Andrew MacGinnitie, Michelle Maciag, Samantha Minniccozzi, Allison E. Norton, Iris M. Otani, Miguel Park, Sarita Patil, Elizabeth J. Phillips, Matthieu Picard, Craig D. Platt, Rima Rachid, Tito Rodriguez, Antonino Romano, Cosby A. Stone, Maria Jose Torres, Miriam Verdú, Alberta L. Wang, Paige Wickner, Anna R. Wolfson, Johnson T. Wong, Christina Yee, Joseph Zhou, Mariana Castells, Practical Guidance for the Evaluation and Management of Drug Hypersensitivity: Specific Drugs, The Journal of Allergy and Clinical Immunology: In Practice, Volume 8, Issue 9, Supplement, 2020, Pages S16-S116,

